

S3.2

P1. Yes

P2. Yes

P3. No.

Let $x = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, x clearly belongs to V , but $2x \notin V$ since $2^2 + 0 + 0 = 4 \neq 1$.

P4. No.

Let $x = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $x \in V$ but $2x \notin V$.

P5. No.

Let $x = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $x \in V$ but $2x \notin V$.

P6. Yes.

$$x_1 + x_2 + x_3 = 0 \quad - (1)$$

$$x_1 - x_2 + 2x_3 = 0 \quad - (2)$$

$$3x_1 - x_2 + 5x_3 = 0 \quad - (3)$$

$$\begin{aligned} (2) - (1) &\Rightarrow -2x_2 + x_3 = 0 \Rightarrow x_3 = 2x_2 \\ (3) - 3 \times (1) &\Rightarrow -4x_2 + 2x_3 = 0 \Rightarrow x_1 = -3x_2 \\ \Rightarrow X &= \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -3x_2 \\ x_2 \\ 2x_2 \end{bmatrix} = x_2 \begin{bmatrix} -3 \\ 1 \\ 2 \end{bmatrix} \end{aligned}$$

P7. Yes

P8. Yes

P9. No.

Let $f(t) = t^3$, $f \in V$, but $2f \notin V$.

P10. No.

Let $y(t)$ be a solution of $y'' + y = \cos t$, i.e. $L[y](t) = \cos t$.

Then $L[2y](t) = 2L[y](t) = 2\cos t$, i.e., $2y(t)$ isn't a solution.

P11. Yes.

P12. Let $x(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix}$ be a solution, i.e. $\frac{dx_1}{dt} = x_2 + 1$ and $\frac{dx_2}{dt} = x_1 + 1$

Consider $y(t) = 2x(t) = \begin{pmatrix} 2x_1(t) \\ 2x_2(t) \end{pmatrix}$.

$$\begin{aligned} \frac{dy_1}{dt} &= \frac{2dx_1}{dt} = 2(x_2 + 1) = y_2 + 2 \\ \frac{dy_2}{dt} &= \frac{2dx_2}{dt} = 2(x_1 + 1) = y_1 + 2 \end{aligned} \Rightarrow y(t) \text{ is not a solution.}$$